



OPTICS (LIGHT) — FULL MIND MAP

प्रकाशिकी · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Light Basics, Spectrum & Speed

- Photon energy $E = hc/\lambda$; wavelength chhoti hogi to energy zyada hogi. Source options me 200–280 nm radiation ka energy per quantum highest hai. [Q1]
- Visible light ko exam me roughly 400–700 nm maana jata hai; older ranges 390–780 nm ya 3900–7600 Å bhi options me mil sakti hain. [Q2–Q4]
- Normal photopic vision me human eye yellow-green region (~555 nm / 5500 Å) ke around sabse sensitive hoti hai; red par maximum sensitivity nahi hoti. [Q5–Q7]
- Light dual nature show karti hai—wave bhi, photon/particle bhi. Everyday scale par straight-line propagation uski very small wavelength ki wajah se apparent hoti hai. [Q8–Q9]
- Vacuum me light speed $c \approx 3 \times 10^8$ m/s hai aur given ordinary objects/waves se bahut zyada hai. [Q10–Q12]
- Medium ki optical density/refractive index badhne par light ki speed kam hoti hai; vacuum me maximum aur given options me glass jaise denser medium me lower hoti hai. Air ki optical properties temperature/pressure/density se subtly vary kar sakti hain. [Q13–Q16]
- Sunlight ko Earth tak pahunchne me ~8.3 min (about 8–8.5 min) lagte hain. [Q17–Q19]
- Light velocity fundamental maximum speed hai; medium change par frequency constant rehti hai, jabki speed aur wavelength change karte hain. Denser → rarer jaane par ray speed up karke normal se away bend karti hai. [Q20–Q24]

⚠ EXAM TRAP — Q13 — No Single Clean Option

► Source itself notes that speed of light in air depends on air's optical density, which changes with temperature, pressure and composition/humidity. Isliye Q13 me single-option framing imperfect hai; concept yaad rakho: $v = c/n$.

2. Interference, Total Internal Reflection & Optical Fibre

- Thin soap film ke glittering colours reflected waves ki interference se bante hain; film ki multiple reflections path difference create karti hain. [Q25]
- Total internal reflection tab possible hai jab light denser se rarer medium me jaye aur incidence angle critical angle se greater ho. Diamond ka high refractive index/low critical angle uski brilliance ka main reason hai. [Q26–Q30]
- Optical fibre total internal reflection par based hota hai; signal lightwave ke form me travel karta hai, power consumption low hota hai aur RF interference se immunity milti hai. [Q31–Q35]
- Medical endoscope me fibre optics/TIR ka use karke body ke internal parts dekhe ja sakte hain. [Q36–Q37]
- Mirage hot surface ke paas refractive-index gradient se ray bending aur suitable conditions me total internal reflection ka result hota hai; terrestrial heating is gradient ko create karti hai. [Q38–Q40]

⚠ EXAM TRAP — Q25 & Mirage — Source Wording vs Core Physics

► Soap-film colours ka core mechanism thin-film interference hai; reflections dono film surfaces se aati hain. 'Total reflection + interference' source option ko verbatim fact mat samjho.

► Mirage ko graded refraction se start karke limiting case me total internal reflection se explain kiya jata hai; source Q40 terrestrial heating + refraction ko select karta hai.

3. Optical Applications — Reflection, Holography, Raman & CD

- Fresh snow ka albedo high hota hai, isliye given surfaces me ye sunlight ka large fraction reflect karta hai. [Q41]
- Three-dimensional image recording/reconstruction technique holography hai. [Q42–Q43]
- Phosphor-coated screen par electron beam ko line-wise scan karke image generate karna raster scan kehlata hai. [Q44]
- Raman effect transparent media se light ke scattering ke dauran frequency shift se related phenomenon hai. [Q45]
- CD audio player information read karne ke liye laser beam use karta hai; CD surface par rainbow-like colours reflection + diffraction/interference effects ki wajah se dikhte hain. [Q46–Q47]
- Radio waves rapidly oscillating electric currents se generate hoti hain; reflected short radio waves ka Doppler effect speed detection me use kiya ja sakta hai. Long wavelength ka scattering short wavelength se more nahi hota, aur water me visible light speed vacuum se noticeably lower hoti hai. [Q48]

4. Human Eye — Retina, Image & Vision

- Cinema me ~ 24 frames/s continuous motion ka illusion deta hai; persistence of vision ko source $\sim 1/16$ s ke order me treat karta hai. [Q49]
- Retina light ko neural impulses me convert karti hai. Eye lens retina par real, inverted aur object se smaller image banata hai. [Q50–Q52]
- Do eyes stereoscopic/binocular vision deti hain, jis se distance aur depth perception improve hota hai. [Q53]
- Solar eclipse ke partial phases ko naked eye se dekhna unsafe hai; source retinal injury ko UV exposure se associate karta hai. [Q54]
- Star ka colour uske surface temperature ka indicator hota hai—bluish stars hotter aur reddish stars cooler hote hain. [Q55]

EXAM TRAP — Q49 & Q54 — Exam-source Specific

- Q49 source persistence of vision ko $\sim 1/16$ s treat karta hai; older exam texts me $1/16$ s ya ~ 0.1 s dono approximations mil sakti hain.
- Q54 source key UV deta hai. Practical eclipse safety point: uneclipsed/partially eclipsed Sun ko certified solar filter ke bina directly kabhi na dekhien.

5. Colour, Wavelength & Atmospheric Optics

- Atmosphere me dust/air particles light ko scatter/diffuse karte hain. Visible colour wavelength se determine hota hai: red longest wavelength, violet shortest. [Q56–Q62]
- Energy wavelength ke inversely proportional hai; given options me blue light highest-energy choice ho sakti hai. Prism me red least deviate karta hai because uska refractive index relatively lower hota hai; red green se less energetic hota hai. [Q63–Q65]
- Atmospheric refraction ki wajah se Sun actual sunrise se pehle visible ho sakta hai, horizon ke paas Sun/Moon flattened lag sakte hain aur stars twinkle karte hain. [Q66–Q68]
- Red danger/traffic signal me useful hai because longest visible wavelength hone se Rayleigh scattering least hoti hai, so long distance/fog me comparatively better penetrate karta hai. [Q69–Q71]
- Sky blue short-wavelength scattering ki wajah se hai. Rainbow ka main cause scattering nahi, balki refraction + dispersion + internal reflection hai; sunrise/sunset red longer atmospheric path me short wavelengths ke scattering-out hone se dikhta hai. [Q72–Q77]
- High altitude/space me atmosphere sparse ya absent hone se scattering kam hoti hai aur sky dark/black dikhta hai. [Q78–Q80]
- Sea ka blue appearance water ke selective absorption/scattering aur sky reflection dono se influenced ho sakta hai; green grass green wavelengths reflect karti hai. [Q81–Q82]
- Soap film ke colours thin-film interference se bante hain. Receding source ka light relativistic Doppler effect se red-shift hota hai; source numerical example blue ko yellow-orange region tak shift karta hai. [Q83–Q84]

⚠ EXAM TRAP — Q81 & Q84 — Simplified Source Explanations

- ▶ Sea blue hona sirf sky reflection nahi hai; water ki wavelength-selective absorption/scattering bhi important hai. Source answer reflection + scattering ko emphasize karta hai.
- ▶ Q84 source classical-style Doppler calculation se yellow-orange answer deta hai. Relativistic speeds par exact treatment relativistic Doppler formula se hota hai; PYQ me source key ko context ke saath padho.

6. Mirrors, Telescope & Reflection Applications

- Convex mirror hamesha virtual, erect aur diminished image deta hai aur wide field of view ke karan vehicle rear-view mirror me use hota hai. [Q85, Q87–Q89]
- Convex lens ke kuch portions black strips se block karne par complete image still ban sakti hai, lekin brightness kam ho jati hai because fewer rays contribute. [Q86]
- Concave mirror me object centre of curvature par ho to image real, inverted, same size banti hai. Pole-focus ke beech object ho to virtual, erect, magnified image mil sakti hai. [Q90–Q92]
- Searchlight/headlight me concave mirror bulb ko focus ke paas rakhkar parallel beam banata hai; dentist enlarged erect view ke liye concave mirror use karta hai. [Q93–Q96]
- Sun protection ke source answer me umbrella top white (reflection) aur bottom black diya gaya hai. [Q97]
- Astronomical telescope final image virtual and magnified de sakta hai; objective diameter badhne se resolving power improve hoti hai. [Q98–Q99]
- Do parallel plane mirrors ke beech ideal case me infinite images form hote hain. Mirror θ angle rotate ho to reflected ray 2θ rotate karta hai. [Q100–Q102]
- Traffic reflectors/glass beads incident light ko strongly return karne ke liye refraction + internal reflection/retroreflection principles exploit karte hain; source TIR ko key mechanism treat karta hai. [Q103]

⚠ EXAM TRAP — Q97 & Q103 — Application Questions

- ▶ Umbrella colour-combination result thermal design details par depend kar sakta hai; source key top white & bottom black deta hai.
- ▶ Traffic glass-bead reflectors ko better description retroreflection hai, jisme refraction/internal reflection combine ho sakte hain; source Q103 TIR ko explanation ke रूप में accept karta hai.

7. Dispersion, Rainbow, Refraction & Plane Mirror

- White light ke seven colours prism se separate hote hain; is splitting ko dispersion kehte hain. [Q104, Q107]
- Rainbow me outer/terminal colours red aur violet hote hain; green middle region me hota hai. Formation me refraction, dispersion aur internal reflection tino contribute karte hain. [Q105–Q109]
- Rainbow observer ke opposite-Sun direction me dikhta hai; high noon me Sun high hone par ordinary ground observer ko primary rainbow generally visible nahi hota. [Q110]
- Shadow light ki rectilinear propagation ki wajah se banta hai. [Q111]
- Pool shallow dikhna, slanted stick bent/short dikhna, aur water me coin ka apparent position shift hona refraction ke examples hain. [Q112–Q114]
- Water ke andar air bubble surrounding water ke comparison me lower refractive index hone se concave/diverging lens ki tarah behave karta hai. Equal focal-length convex + concave lenses contact me net power ~ 0 dekar plane glass sheet jaisa act kar sakte hain. [Q115–Q117]
- Full-body image dekhne ke liye plane mirror ki minimum height person ki height ka half hoti hai; plane mirror ka radius of curvature infinity hota hai. [Q118–Q120]

8. Lenses, Eye Defects & Accommodation

- Magnifying glass convex lens hota hai. Hypermetropia/long-sightedness convex lens se correct hoti hai; reading glasses commonly converging/convex lenses use karte hain. [Q121–Q124]
- Myopia/short-sightedness concave lens se correct hoti hai; combined rule: Myopia→concave, Hypermetropia→convex. [Q125–Q129]
- Bright se dark room me jaane par initial vision poor hoti hai because pupil dilation aur rod pigment (rhodopsin) regeneration/dark adaptation me time lagta hai. [Q130]

- Eye focusing accommodation se hoti hai: ciliary muscles lens curvature/convexity change karke focal length adjust karte hain. [Q131–Q132]
- ENT examination ke traditional head mirror me concave mirror with central aperture use hota hai, convex nahi. [Q133]
- Normal eye ka least distance of distinct vision ~ 25 cm hota hai. Lens power diopetre me measure hoti hai: $P = 1/f(m)$; +2 D lens ka focal length $+0.5$ m = 50 cm hota hai. [Q134–Q139]
- Ordinary non-corrective sunglasses ka optical power approximately 0 diopetre hota hai; unka main role light intensity reduce karna hai. [Q140]

EXAM TRAP — Q130 — Multiple Mechanisms in Dark Adaptation

► Dark room me adaptation sirf pupil-size change se nahi hoti; rod photopigment (rhodopsin) regeneration aur neural adaptation bhi role play karte hain. Source ne isliye single clean correct option mark nahi kiya.

9. Interference, Camera & Retina Comparison

- Young double-slit fringe width $\beta = \lambda D/d$. $\lambda = 500$ nm, $D = 1$ m, $d = 1$ mm ke liye $\beta = 0.5$ mm. [Q141]
- Camera exposure roughly aperture area $\times$ shutter time par depend karta hai; given combinations me f/8 at 1/250 s maximum exposure deta hai. [Q142]
- Traditional camera me retina film ke analogous hai; normal human eye me object image retina par form hoti hai. [Q143–Q145]
- Hypermetropia me near object ki image retina ke behind form hone ki tendency hoti hai. [Q146]

10. Colour Mixing, Pigments & Optical Illusions

- Opaque object ka observed colour mainly us wavelength/colour se determined hota hai jise wo reflect karta hai. [Q147]
- Additive primary colours of light Red, Green, Blue (RGB) hain. In tino ko equal proportion me combine karne par white light banti hai; black primary colour nahi hai. [Q148–Q152]
- Subtractive pigment system me cyan, magenta, yellow important hain: magenta+yellow $\rightarrow$ red aur cyan+magenta $\rightarrow$ blue. Source matching: Magenta=red+blue; Teal=blue+green+white; Mauve=blue+red+white; Cyan=green+blue. [Q153–Q154]
- Sunlight/white light ko school-level spectrum me seven VIBGYOR colours me describe kiya jata hai; brown VIBGYOR member nahi hai. [Q155–Q156]
- Source optical-illusion set me dusk par Sun ka apparent size, dawn ka red colour aur star twinkling ko optical effects/illusions maana gaya hai; dawn me Moon ya Polestar visibility real astronomical observations hain. [Q157]